

UNIT 10

GEOMETRY



LINE SEGMENT

Segment means a part. Line segment is a part of a line. It has two distinct end points. In the following figure we can see a line L with points P and Q marked on it.



The differences between line, line segment and Ray:

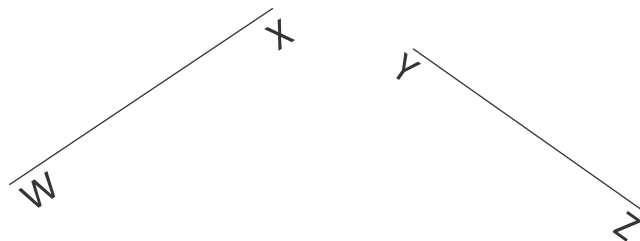
With respect to	Line	Line Segment	Ray
End points	No end point	Two end points	One end point
Length	No definite length	Definite length	No definite length
Symbol	Line AB or $\overleftrightarrow{AB}$	Line segment AB or $\overline{AB}$	Ray AB or $\overrightarrow{AB}$

Add measures of two or more line segments

Adding measures of two or more line segments, means to find another line segment whose length is the sum of measures of two given line segments.

Example: Add the given line segments $m\overline{WX} = 4$ cm and $m\overline{YZ} = 3$ cm.

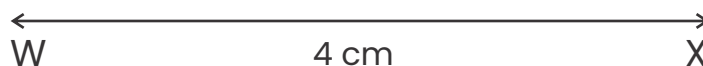
Using pair of compasses, pencil and ruler.



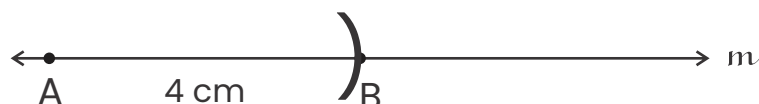
1. Draw a line m of any suitable length and mark a point A on it.



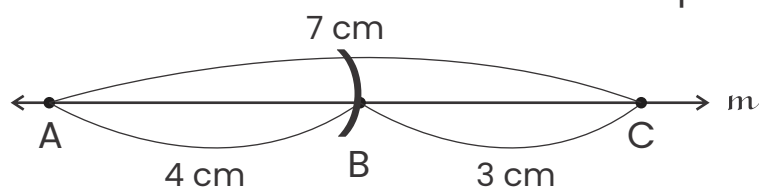
2. Use pair of compasses to measure the length of $WX = 4$ cm.



3. Keep the radius equals to 4 cm, with centre at point A, draw an arc to cut line m at point B.



4. Repeat step 2 to get radius equal to 3 cm and with centre B, draw an arc to cut line m in the same direction at point C.

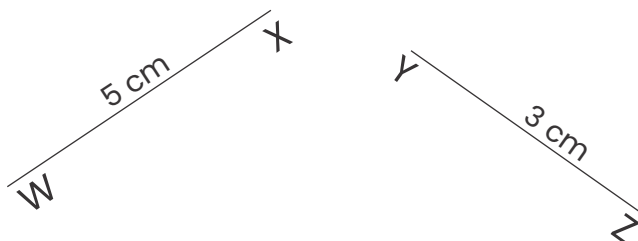


Therefore, $m\overline{AC} = m\overline{WX} + m\overline{YZ} = 4 \text{ cm} + 3 \text{ cm} = 7 \text{ cm}$

Subtract measure of line segment from a longer one:

For subtracting measure of one line segment from a longer one, we are going to find another line segment whose measure is the difference of measures of two given line segments.

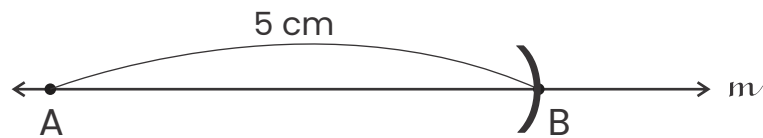
Example: Subtract the measure of line segment YZ from the measure of line segment WX . Using pair of compasses, pencil and ruler.



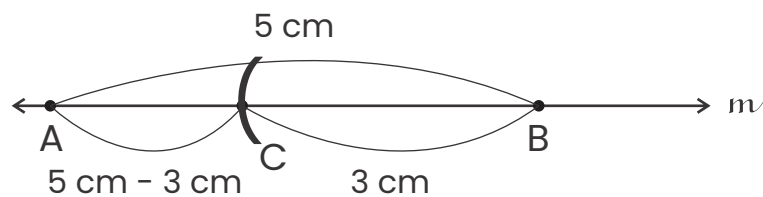
1. Draw a line m of any suitable length and mark a point A on it.



2. Measure the length of $\overline{WX}$ with centre A, draw an arc to cut line m at point B.



3. With centre B and radius equal to $\overline{YZ}$, draw another arc to cut line m at point C in the opposite direction.

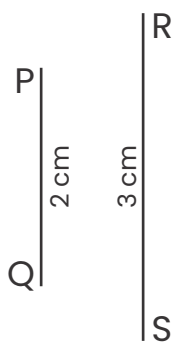


Therefore, $m\overline{AC} = m\overline{WX} - m\overline{YZ} = 5 \text{ cm} - 3 \text{ cm} = 2 \text{ cm}$

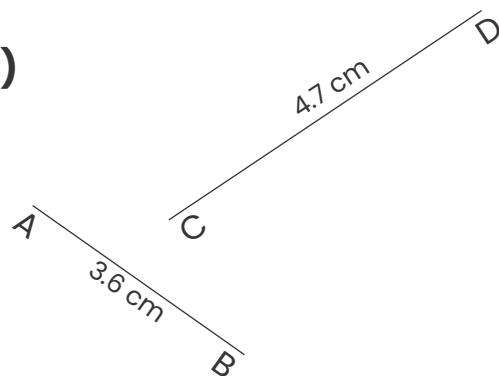
EXERCISE 10.1

Q1. Find the sum of measures of the following pairs of line segments geometrically.

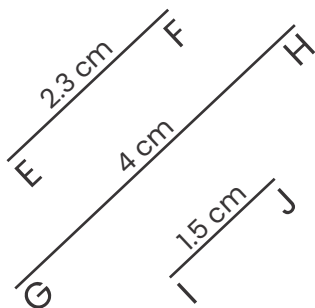
(i)



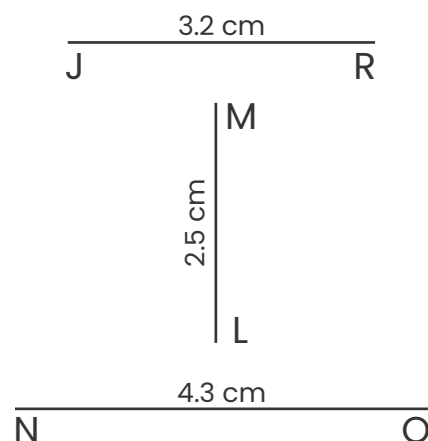
(ii)



(iii)

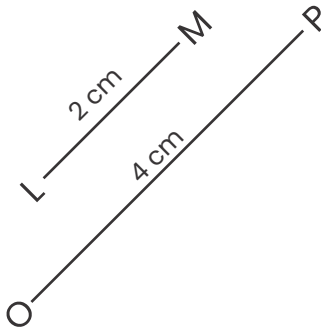


(iv)



Q2. Find the difference of measures of the following line segments geometrically. Subtract shorter length from longer length.

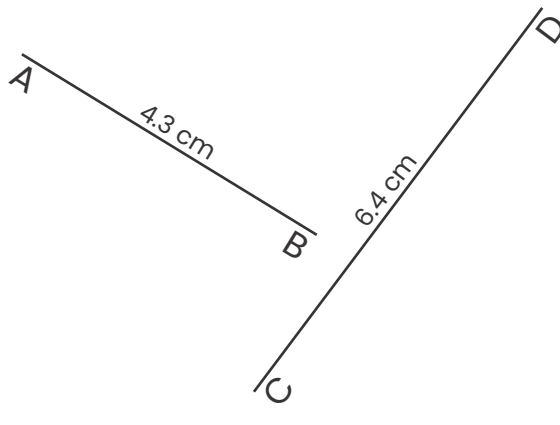
(i)



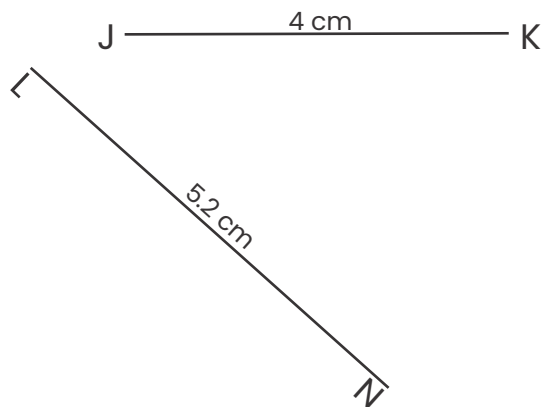
(ii)



(iii)

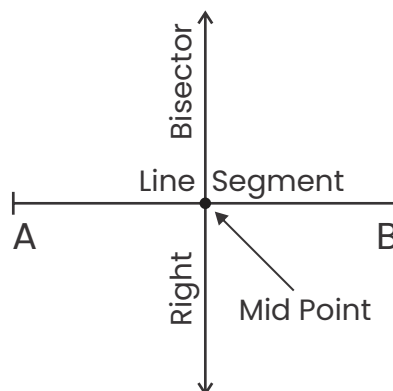


(iv)



RIGHT BISECTOR

A line which makes an angle of 90° with the given line segment and divide it into two equal parts is called right bisector.



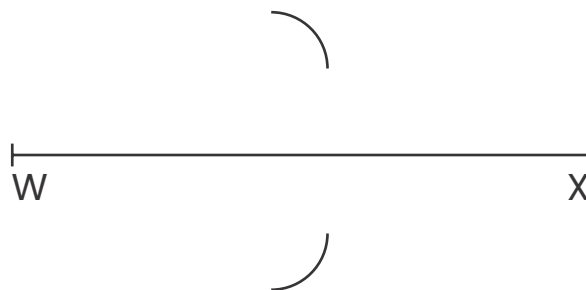
Example: Draw a right bisector of $\overline{WX}$ measuring 6 cm by using pair of compasses. Consider the given line segment WX measuring 6 cm.



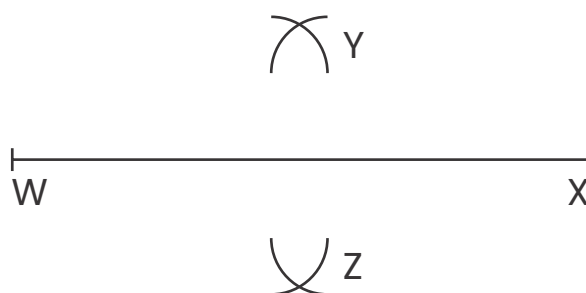
Now follow the following steps by using pair of compasses, pencil and ruler.

Steps of construction:

Step I: With centre W and radius more than half of $\overline{WX}$ (say 3.5 cm), draw two arcs on each side of $\overline{WX}$.

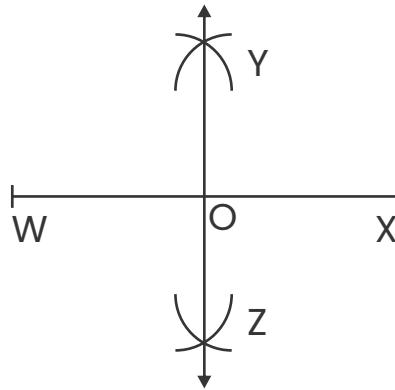


Step II: With centre X draw two arcs of same radius to cut previous arcs at points Y and Z.



Step III: Draw $\overline{YZ}$ which cuts $\overline{WX}$ at point O. Therefore $\overline{YZ}$ bisects the given line segment WX at point O. as $m\overline{WO} = m\overline{OX}$.

Hence $\overline{YZ}$ is the required right bisector of $\overline{WX}$.



$\overleftrightarrow{OY}$ and $\overleftrightarrow{OZ}$ are also right bisectors of $\overline{WX}$.

Draw a perpendicular to a given line from a point on it using compasses:

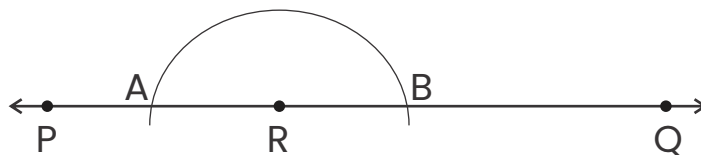
Example: Draw a perpendicular to the line PQ at point R on it by using pair of compasses.

Steps of construction:

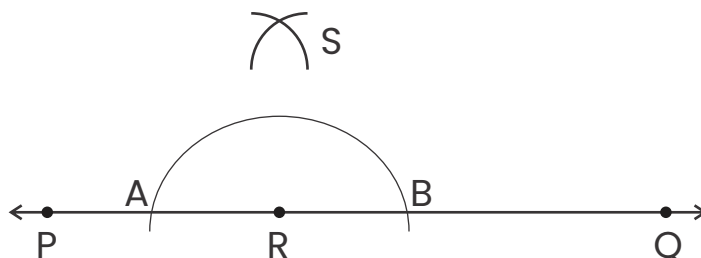
Step I: Draw a line PQ, mark point R on it.



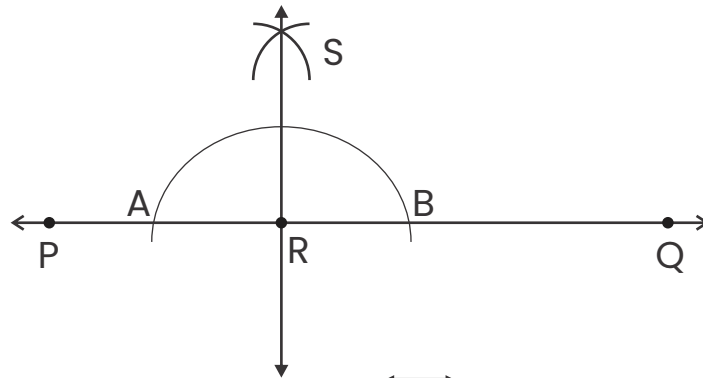
Step II: With centre R draw an arc of suitable radius to cut PQ at points A and B, such that $m\overline{RA} = m\overline{RB}$.



Step III: With centres A and B draw two arcs of radius more than $m\overline{RA}$, to cut each other at point S.



Step IV: Join point R and S, and extend it on both sides of line $\overleftrightarrow{PQ}$ to form $\overleftrightarrow{RS}$.



$\overleftrightarrow{RS}$ is the required perpendicular to $\overleftrightarrow{PQ}$ from a point R on it.

EXERCISES 10.2

Q1. Draw right bisectors of the line segments of the following measures.

(i) 5cm

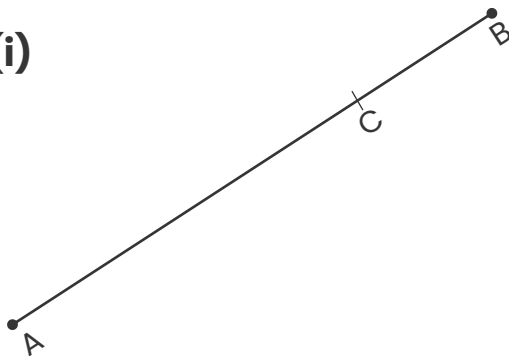
(ii) 6.3cm

(iii) 8cm

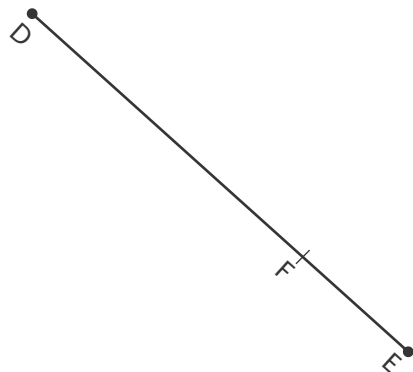
(iv) 6cm

Q2. Draw perpendicular from given points on the following line segments.

(i)



(ii)



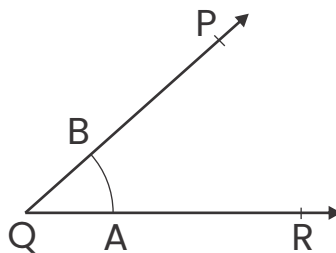
(i)



CONSTRUCTION OF ANGLES

Use compasses to construct an angle equal in measure of a given angle:

Example: Construct an angle equal in measure of the angle PQR. Consider the given angle PQR.



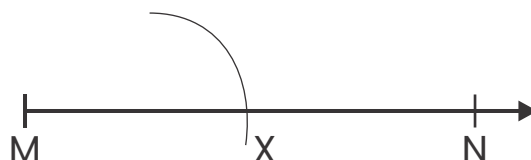
Steps of construction:

Step I: Draw MN.

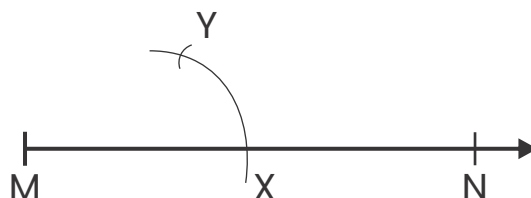


Step II: With center Q draw an arc of any suitable radius, to cut $\overrightarrow{QR}$ and $\overrightarrow{QP}$ at points A and B respectively.

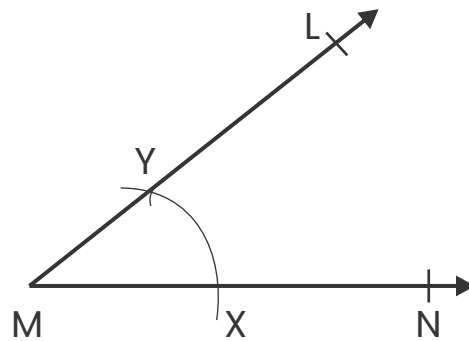
Step III: With centre M, draw an arc of radius equals to $\overline{QA}$ to cut $\overrightarrow{MN}$ at point X.



Step IV: Now place the needle of the compass on A and open the arm till radius AB. With centre X and taking the same radius draw an arc to cut the previous arc at point Y.



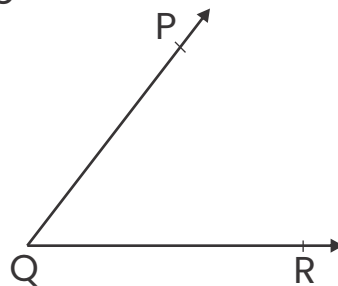
Step V: Join point M with Y and extend to the point L through point Y. Measure it with protractor as $m\angle LMN = m\angle PQR$.



Hence LMN is the required angle.

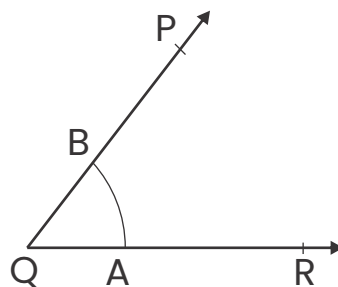
Use compasses to bisect a given angle:

Example: Draw the bisector of the angle PQR.
Consider the given angle PQR.

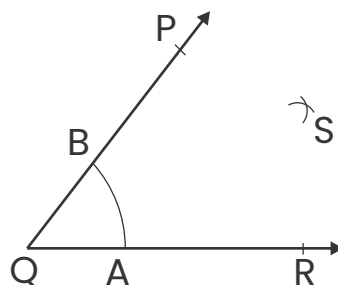


Steps of construction:

Step I: With centre Q draw an arc of any suitable radius to cut $\overrightarrow{QR}$ and $\overrightarrow{QP}$ at points A and B respectively.

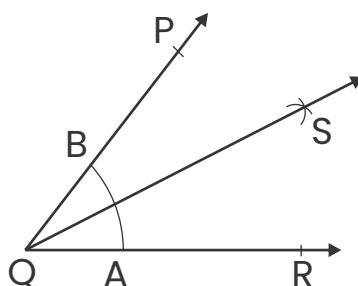


Step II: With centres A and B draw two arcs of suitable radius to cut each other at point S.



Step III: S Join point Q with S and draw QS.

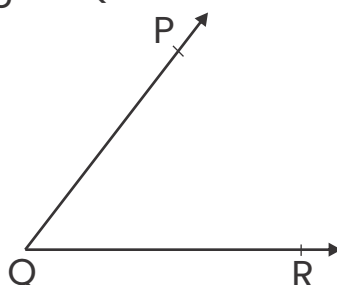
Here QS bisects PQR in two equal angles i.e $m\angle PQS = m\angle RQS$



Use compasses to divide a given angle into four equal angles:

Example: Divide the angle PQR into four equal angles.

Consider the given angle PQR.



Steps of construction:

Step I: First of all bisect the given angle PQR in two equal angles PQS and RQS using compass as shown in the figure

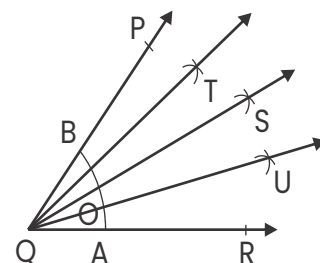
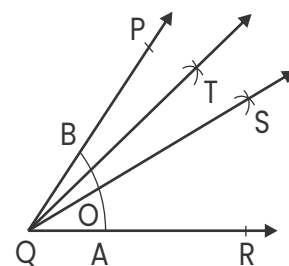
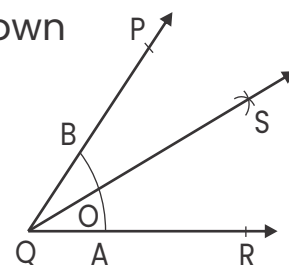
Step II: Name the intersecting point of QS as O.

Step III: With centres O and B and taking any suitable radius, draw two arcs to cut each other at point T. Join Q with T to get QT.

Step IV: With centres O and A, repeat the same process to get QU.

Now the given PQR has been divided into four equal angles.

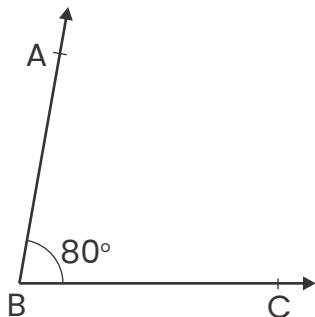
i.e $m\angle PQT = m\angle TQS = m\angle SQU = m\angle UQR$



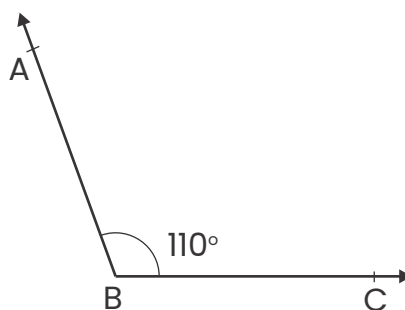
EXERCISES 10.3

Q1. Bisect the following angles with the help of compasses.

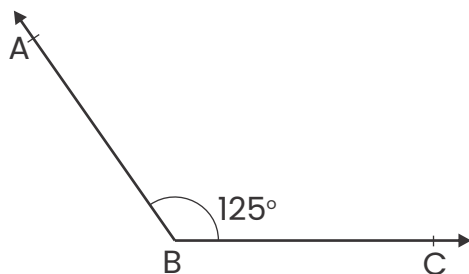
(I)



(II)

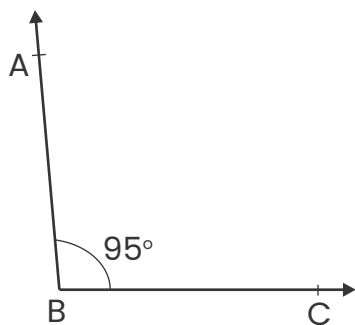


(III)

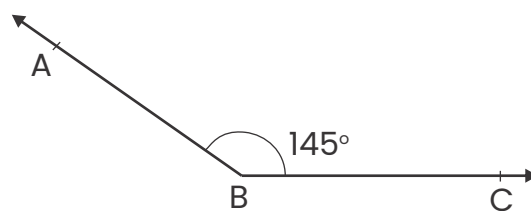


Q1. Bisect the following angles with the help of compasses.

(I)



(II)



TRIANGLES

Triangle is a plane closed geometrical figure bounded by three sides. It has three sides and three angles.

CONSTRUCTION OF TRIANGLES

Construct a triangle when three sides (SSS) are given:

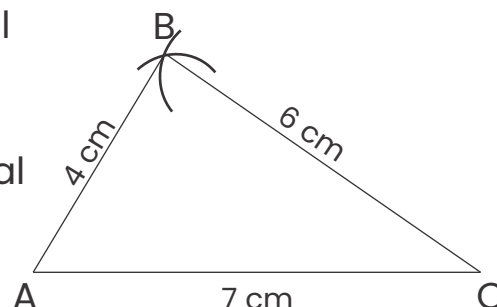
Example: Construct a triangle ABC when $m\overline{AB} = 4\text{ cm}$, $m\overline{BC} = 6\text{ cm}$ and $m\overline{AC} = 7\text{ cm}$.

Steps of construction:

Step I: Draw $m\overline{AC} = 7\text{ cm}$.

Step II: With centre A and radius equal to 4 cm, draw an arc.

Step III: With centre C and radius equal to 6 cm, draw another arc to cut the previous arc at point B.



Can we construct a triangle with 3 sides measures 4cm, 3cm, and 2 cm?

Step IV: Join point B to A and C.
The ABC is constructed.

Construct a triangle when two sides and their included angle (SAS) are given:

Example: Construct a triangle DEF when $m\overline{DE} = 6\text{ cm}$, $m\overline{DF} = 4\text{ cm}$ and $m\angle D = 70^\circ$.

Steps of construction:

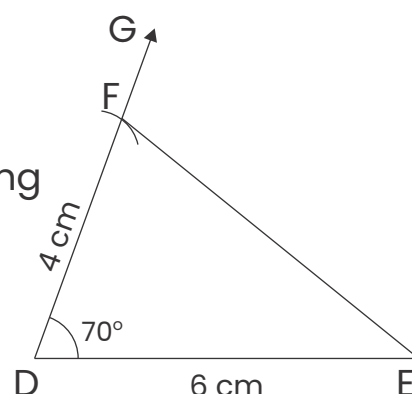
Step I: Draw $m\overline{DE} = 6\text{ cm}$.

Step II: At point D draw $m\angle EDG = 70^\circ$ by using protractor.

Step III: With centre D and radius equal to 4 cm, draw an arc to cut $\overrightarrow{DG}$ at point F.

Step IV: Join point F with E.

Hence, the required DEF is constructed.



Construct a triangle when two angles and included side (ASA) are given:

Example: Construct a triangle MNO when $m\overline{NO} = 5\text{ cm}$, $m\angle N = 60^\circ$ and $m\angle O = 55^\circ$.

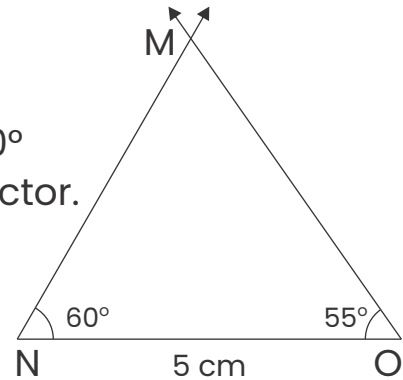
Steps of construction:

Step I: Draw $m\overline{NO} = 5\text{ cm}$.

Step II: At points N and O, draw angles of 60° and 55° respectively with the help of protractor. Name the point where they intersect each other.

Step III: The arms of both angles intersect at point M.

Hence, the required $\triangle MNO$ is constructed.



Construct a triangle when hypotenuse and one side (RHS) for a right angled triangle are given:

Example: Construct a right angled triangle PQR when hypotenuse $\overline{PQ}$ is 7 cm long and a side $\overline{QR}$ is 5 cm long and $m\angle R = 90^\circ$.

Steps of construction:

Step I: Draw $\overrightarrow{RA}$ greater than 7 cm.

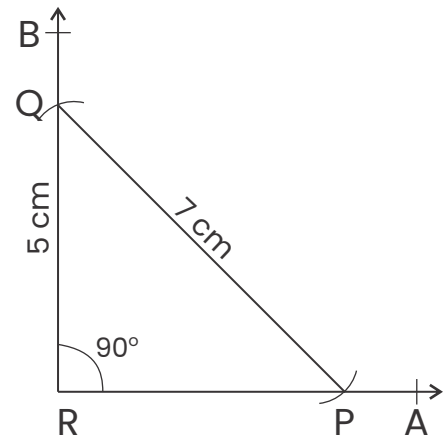
Step II: At point R draw $m\angle ARB = 90^\circ$.

Step III: With centre R and radius equal to 5 cm, draw an arc to cut $\overrightarrow{RB}$ at point Q.

Step IV: With centre Q and radius equal to 7 cm, draw an arc to cut $\overrightarrow{RA}$ at point P.

Step V: Join point P and Q.

Hence, the required $\triangle PQR$ is constructed and it is the right angled triangle.



EXERCISES 10.4

Q1. Construct a triangle ABC when:

- (i) $m\overline{AB} = 3\text{cm}$, $m\overline{BC} = 5\text{cm}$, and $m\overline{CA} = 7\text{cm}$.
- (ii) $m\overline{AB} = 2\text{cm}$, $m\overline{BC} = 5\text{cm}$, and $m\overline{CA} = 5\text{cm}$.
- (iii) $m\overline{AB} = 5\text{cm}$, $m\overline{BC} = 7\text{cm}$, and $m\overline{CA} = 9\text{cm}$.
- (iv) $m\overline{AB} = 8\text{cm}$, $m\overline{BC} = 6\text{cm}$, and $m\overline{CA} = 10\text{cm}$.

Q2. Construct a triangle MNO when:

- (i) $m\angle N = 63^\circ$, $m\overline{NO} = 3\text{cm}$, and $m\overline{MN} = 5\text{cm}$.
- (ii) $m\angle N = 75^\circ$, $m\overline{NO} = 3.4\text{cm}$, and $m\overline{MN} = 4\text{cm}$.

Q3. Construct a right angled triangle XYZ when $m\angle Z = 90^\circ$.

- (i) $m\overline{XY} = 7\text{cm}$, $m\overline{XZ} = 5\text{cm}$.
- (ii) $m\overline{XY} = 10\text{cm}$, $m\overline{XZ} = 6\text{cm}$.

Q4. Construct a triangle ABC according to the following conditions and write steps of construction in each case.

- (i) $m\overline{AB} = 8\text{cm}$, $m\overline{AC} = 6\text{cm}$, and $m\angle A = 65^\circ$.
- (ii) $m\angle A = 45^\circ$, $m\angle C = 45^\circ$, and $m\overline{AC} = 7.5\text{cm}$.
- (iii) $m\overline{AB} = 5\text{cm}$, $m\overline{AC} = 7\text{cm}$, and $m\angle B = 90^\circ$.